Report for the project on Ramsey property and MAD families

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Contents

In this report, I summarize the the project on Ellentuck space, Ramsey property of set of reals and infinite mad families. The material I studied covers the relevant section in Chapter 26 of Jech [Jec00], the paper [Tör18] and [ST19]. In the following, Jech refers to [Jec00] and Halbeisen stands for [Hal17].

1 Ellentuck Space and Mathias Forcing

In this chapter, I study the basic properties of Ellentuck space, Ramsey set of reals and Mathias Forcing following the section *Ramsey Sets of Reals and Mathias Forcing* in Ch. 26 of Jech and relevant chapters in Halbeisen, especially chapter 10.

Recall the following notation in Ramsey theory:

$$\kappa \to (\lambda)^{\alpha}_{\beta}$$

Which means that every partition of $[\kappa]^{\alpha}$ into β sets has a λ sized homogenous set.

Given axiom of choice, the following gives an counterexample to $\omega \to (\omega)^{\omega}_{2}$.

Theorem 1. *(AC)* $\omega \not\to (\omega)^{\omega}_{2}$.

Proof. Consider the following equivalence relation of $[\omega]^{\omega}$:

$$x \sim y \iff x \Delta y \text{ is finite.}$$

Then let A be a representative set for the equivlence relation $\sim$. (Note: this is not necessarily a mad family, as representatives may be almost disjoint) Consider the following colouring: $\pi(x) = 1$ iff $|x\Delta a|$ is even where $x \in [a]$ and $a \in A$.

Then for any infinite set $X \in [\omega]^{\omega}$, $[X]^{\omega}$ is not homogeneous. □

Ramsey property is a regularity property linked to the above result. A set $R \subseteq [\omega]^{\omega}$ is *Ramsey* if there is an infinite homogeneous set, i.e. there is $X \in [\omega]^{\omega}$ s.t. $[X]^{\omega} \subseteq R$ or $R \cap [X]^{\omega} = \emptyset$.

Here, a subset of $[\omega]^{\omega}$ is understood as a 2-color coloring. A set is Ramsey if it is a coloring that is not pathological in the sense of theorem 1.

2 Ellentuck topology

The Ellentuck topology is the topology on $[\omega]^{\omega}$ generated by basic open sets of the form

$$[s, H]^{\omega} = \{x \in [\omega]^{\omega} \mid s \subseteq x, x - max(s) \in H\}$$

Where $s \in \omega^{<\omega}$ and $H \in [\omega]^{\omega}$. From now on, we write $x - max(s)$ as x/s.

2.1 Ellentuck topology vs the subspace topology

How is this connected to the usual topology on the real? $[\omega]^{\omega}$ can be viewed as a $\mathbf{\Pi}^0_1$ subset of 2^{ω}, as $\forall n \in \omega \exists n > m, x(n) = 1$ is a $\mathbf{\Pi}^0_1$ property. A set open in the subspace topology is open in Ellentuck topology, but the converse is not true. But it turns out that every basic open in Ellentuck topology is closed for the subspace topology:

$[s, H]^{\omega}$ can be expressed as $[\omega]^{\omega} \cap \{x \mid s \subseteq x\} \cap \bigcap_{n \notin H} \{x \mid x(n) = 0\}$.

This also shows that a baisc open in the Ellentuck topology is clopen in the Ellentuck topology.

The usual metric of subsapce topology does not generate the Ellentuck topology. We first observe that there is a meager set in the subspace topology that is considered open dense in the Ellentuck.

Proposition 1 (Proposition 2 in on [Ple87])**.** *There is a meager set in the subspace topology that is considered open dense in the Ellentuck.*

Proof. Let E be the set of even numbers and O be the odds, consider

$$U = \bigcup\{[\emptyset, E \cup x] \mid x \in [\omega]^{<\omega}\} \cup \bigcup\{[\emptyset, O \cup x] \mid x \in [\omega]^{<\omega}\}$$

Each single $\{[\emptyset, E \cup x] \mid x \in [\omega - \bigcup s]^{<\omega}\}$ as for any open $N_s = \{x \mid s \subseteq s\}$ for $s^{<\omega}$, there is always a basic open included in N_s that's disjoint from $[\emptyset, E \mid x]$.

The set is dense in Ellentuck topology, as for basic open $[s, H]$, either $H \cap E$ or $H \cap O$ is infinite, then $[s, H] \cap [\emptyset, E \cap s] \neq \emptyset$ if $E \cap H$ is infinite. □

indeed, the Ellentuck topology is not metritizable, by the following argument.

A topology is normal iff for every pair of disjoint closed set C_1, C_2, there is disjoint open set s.t. $U_1 \supseteq C_1$ and $U_2 \supseteq C_2$. Every metric space is normal.

Theorem 2. *Ellentuck space is not normal, thus does not admit a metric.*

read on completely ramsey sets

Proof. Let h be a bijection from ω^2 to $2\mathbb{N}$, the even numbers. For $A \subseteq \omega$, set $A^* = \{h(t, s) \mid t \in A, s \notin A\}$.

Let O be the set of odd numbers, consider

$$H = \{O \cup A^* \mid |A| = \omega\}, K = \{O \cup A^* \mid |A| < \omega\}$$

The two sets are closed and discrete, to see that they are closed,

$$[\omega]^\omega - H = \bigcup\{[\emptyset, (O \cup A^*) - \{z\}] \mid |A| < \omega, z \in A^* \cup O\} \cup \bigcup\{[\{h(t, s), h(r, t)\}, \mathbb{N}] \mid s, t, r \in \mathbb{N}\}$$

To see this is the case, for $O \cup A^* \in H$, if $A = B$, $O \cup A^* \notin [\emptyset, (O \cup B^*)/\{z\}]$ is obvious, if $A \neq B$, then $A^* - B^*$ and $B^* - A^*$ are both non-empty, which means that $O \cup A^* \notin [\emptyset, O \cup (B^*/\{z\})]$. Moreover, A^* has no pair of sets of the form $\{h(t, s), h(r, t)\}$, hence $O \cup A^*$ is not in any of the latter part either.

On the other hand, if $B - O$ is not of the form A^* for any A, then either O contains a pair $\{h(t, s), h(r, t)\}$, or it is a subset of A^*, where $A = \{s \mid h(s, t) \in B - O\}$, then $B \in O \cup A^* - \{z\}$ for some z.

The set H is discrete as $H \cap [\emptyset, V \cup A^*] = \{V \cup A^*\}$, for $|A| < \omega$.

Similarly, K is also discrete and closed.

We can directly show that there is no open neighbourhoods of H, K repectively that are disjoint, like in [1]

Alternatively, it turns out that $C = \{x \in [\omega]^\omega \mid O \subseteq x\}$ is a separable closed subset, and thus $\{f : C \to \mathbb{R} \mid f \text{ is continuous}\}$ is of size at most 2^ω. Yet by Tietze extension theorem, every continous $g : H \to \mathbb{R}$ extends to $f' : C \to \mathbb{R}$, thus there are at least $2^{|H|}$ many continouse functions $f' : C \to \mathbb{R}$, this is a contradiction. □

Theorem 3 (Tietze extension theorem)**.** *If X is normal, $A \subseteq X$ is closed and $f : A \to \mathbb{R}$ is continuous, then f can be extended to $f' : X \to \mathbb{R}$ s.t.*

[1] https://math.stackexchange.com/questions/1646738/what-are-the-two-disjoint-closed-sets-that-cannot-be-separated-by-two-disjoint-o

The proof idea generalized to the following lemma:[2]

Lemma 1 (Jone's Lemma). *If X is a separable and normal space, and $D \subseteq X$ is a closed set s.t. the subspace topology is discrete, then $2^{|D|} \leq 2^{\omega}$*

Finally, the Ellentuck space is not separable, for arbitrary sequence $x_0, x_1 \dots$ of elements in $[\omega]^{\omega}$, pick $n_i \in x_i$ s.t. $n_i > 1 + n_{i-1}$. The sequence can always be chosen as x_i are infinite. Then let $A = \omega - \{n_i \mid i \in \omega\}$, this is an infinite set as it contains $n_i + 1$ for all i. Then $[\emptyset, A]$ is a basic open that avoids all x_i.

2.2 Ellentuck's Theorem

In what follows we always refer to the Ellentuck topology unless explicitly stated.

Definition 1. *A set $S \subseteq [\omega]^{\omega}$ is completely Ramsey if for every (s, A) there exists an infinite $H \subseteq A$ such that either $[s, H] \subseteq S$ or $[s, H] \cap S = \emptyset$.*

A set $S \subseteq [\omega]^{\omega}$ is Ramsey null if for every (s, A) there exists an infinite $H \subseteq A$ such that $[s, H] \cap S = \emptyset$.

In this subsection we aim at the following result:

Theorem 4 (Ellentuck's Theorem, Lemma 26.27 in Jech). *A set $S \subseteq [\omega]^{\omega}$ is completely Ramsey if and only if it has the Baire property.*

A set $S \subseteq [\omega]^{\omega}$ is Ramsey null iff it is meager iff it is no where dense.

Let $S \subseteq [\omega]^{\omega}$, we say A accepts s if $[s, A] \subseteq S$ and A rejects s if no $X \subseteq A$ accepts s.

Lemma 2 (Lemma 26.28 in Jech). *For any $S \subseteq [\omega]^{\omega}$, there is an X that accepts or rejects each of its finite subset.*

Proof. We construct the set inductively, let X_0 be such that X_0 either accepts or rejects $\emptyset$. The set exists as if all $X_0 \subseteq \omega$ does not accept $\emptyset$, then ω rejects it.

Then let $a_0 = minX_0$ and let $X_1 \subseteq X_0$ be such that it either accepts or rejects $\{a_0\}$. Continue this process and $\{a_0, a_1 \dots\}$ accepts or rejects each of its finite subset as $\{a_n, a_{n+1} \dots\} \subseteq X_{n+1}$ □

Lemma 3 (Lemma 26.29). *For any $S \subseteq [\omega]^{\omega}$, there is a Y that either accepts $\emptyset$ or rejects each of its finite subsets.*

Proof. Let X satisfy the condition in 2. If there is a subset $Y \subseteq X$ s.t. Y accepts $\emptyset$ we are done.

So assume X rejects $\emptyset$, recursively construct $\{a_0, a_1 \dots\} \subseteq X$. Say every subset $\{a_0 \dots a_n\} \subseteq X$ is rejected by X, for each $s \subseteq \{a_0 \dots a_n\}$ there are only finitely many $z \in X$ s.t. X accepts $s \cup \{z\}$. This is because otherwise let Z be the infinite collection of z s.t. that X accepts $s \cup \{z\}$, then Z accepts s as $[s, Z] = [s \cup \{z\}, Z] = [s \cup \{z\}, X]$, this is a contradiction to X rejects s.

Then we can find $a_{n+1} \in X$ s.t. X rejects every subset of $\{a_1 \dots a_{n+1}\}$.

The set $\{a_i \mid i \in \omega\}$ rejects each of its finite subsets as X does. □

[2] https://dantopology.wordpress.com/2012/07/31/jones-lemma/

Lemma 4 (Lemma 26.30)**.** *If U is open, U is Ramsey.*

Proof. Let Y satisfy the conditions in 3. If Y accepts $\emptyset$, then $[S]^\omega = [\emptyset, S] \subseteq U$.

If Y rejects each of its finite subsets, assume for contradiction that $[Y]^\omega \cap S \neq \emptyset$, pick x in the intersection, let $Y \in [s, H] \subseteq U$ by openness, then $Y \in [s, Y - max(s)] \subseteq [s, H]$, this is a contradiction to Y rejects s. □

Lemma 5 (Lemma 26.31)**.** *If U is open, U is completely Ramsey.*

Proof. For arbitrary (s, A), we fix an increasing enumeration $f : \omega \to A$ of A. Consider the following map:

$$f^* : [\omega]^\omega \to [\omega]^\omega$$
$$X \mapsto s \cup f[X]$$

This is a consituous map with respect to the Ellentuck topology: for any $[t, H]$ either s is incompatible with t and the reflection is empty, or the reflection would be open.

Now the set $(f^*)^{-1}[U]$ is open, and hence by the previous Lemma is Ramsey, let $H \in [\omega]^\omega$ be a homogeneous set for $f^{-1}[U]$, then $f^*[H]$ would be $[s, f[H]]$, which satisfies the required condition. □

Remark: It is obvious that both Ramseyness and completely Ramseyness are closed under complements.

Lemma 6 (Lemma 26.32 and 26.33)**.** *Every nowhere dense set is Ramsey null. Ramsey nullness is closed under countable union.*

Proof. We may assume that N is closed and nowhere dense, hence it is completely Ramsey. But since it cannot contain a basic open, it is Ramsey null.

Let $S = \bigcup_{n \in \omega} S_n$ where S_n is Ramsey null, recursively construct $X_0 \supseteq X_1 \supseteq \ldots X_n$ s.t. $X_i \subseteq A$, and $\{a_0 \ldots a_n\} \subseteq A$ s.t. $[t, X_i] \cap S_i = \emptyset$ for any $s \subseteq t \subseteq s \cup \{a_0 \ldots a_n\}$.

Then, let $H = \{a_0 \ldots\}$ and $[s, H] \cap S$ is empty. □

Proof of theorem 4. The second statement follows from the above lemma.

For the first statement, if B has Baire property, write it as $U \Delta M$, then for any (s, A), there is $X \subseteq A$ s.t. $[s, X] \cap M = \emptyset$ and there is $H \subseteq X$ s.t. $[s, X] \cap U = \emptyset$ or $[s, X] \subseteq U$, then $[s, X] \cap B = \emptyset$ or $[s, X] \subseteq B$.

For the other direction, let C be completely Ramsey, consider $U = \bigcup\{[s, H] \mid [s, H] \subseteq C\}$, which is a open subset of C. It suffice to show that $C - U$ is meager. But for arbitrary (s, A), there is $H \subseteq A$ s.t. $[s, H] \subseteq C$, notice $[s, H] \subseteq U$ and hence $[s, H] \cap C - O$ is empty. This shows that $C - U$ is Ramsey null, thus meager. □

3 Mathias Forcing

Definition 2 (Definition 26.24 Mathias Forcing)**.** *In Mathias forcing, a forcing condition is a pair (s, A), $s \in \omega^{<\omega}$ and $A \in [\omega]^\omega$ s.t. $max(s) < min(A)$. We say $(s, A) < (t, B)$ if:*

1. $s \supseteq t$.
2. $A \subseteq B$.
3. $s - t \subseteq B$.

3.1 Pure decision of Mathias Forcing

Lemma 7 (Lemma 26.34)**.** *Let σ be a sentence of the forcing language and let (s, A) be a condition. Then there exists an infinite set $B \subseteq A$ such that (s, B) decides σ.*

Proof. Let $Q^+ = \{p \mid p \Vdash \sigma\}$ and $Q^- = \{p \mid p \Vdash \neg\sigma\}$. $S^+ = \bigcup\{[s, H] \mid (s, H) \in Q^+\}$ and $S^- = \bigcup\{[s, H] \mid (s, H) \in Q^-\}$. Then $S^+ \cup S^-$ is open dense and its complement is nowhere dense in the Ellentuck topology. Pick $B \subseteq A$ s.t. $[s, B] \subseteq S^+ \cup S^-$, further pick $C \subseteq B$ s.t. $[s, C] \subseteq S^+$ or $[s, C] \cap S^+ = \emptyset$, in which case $[s, C] \subseteq S^-$. This means either $(s, C) \in Q^+$ or $(s, C) \in Q^-$, as Q^+ or Q^- would be dense under (s, C). $\square$

3.2 A characterization of Mathias reals

For a generic filter G over Mathias forcing $\mathbb{P}$, it uniquely determines a real x_G by

$$x_G = \bigcup\{s \mid (s, A) \in G \text{ of some } A\}$$

Conversely the filter is determined by x as

$$G = G_x = \{(s, A) \mid s \subseteq x \subseteq x \cup A\}$$

We aim for the following characteristic theorem for Mathias reals.

Theorem 5. *[AC] Let M be a transitive model of ZFC. An infinite set $x \subseteq \omega$ is a Mathias real over M if and only if for every maximal almost disjoint family $\mathcal{A} \in M$ of subsets of ω, there exists an $X \in \mathcal{A}$ such that $x - X$ is finite.*

For D an open dense subset of Mathias forcing, for $X \in [\omega]^\omega$ and $s \in [\omega]^{<\omega}$ s.t. $maxs < minX$, we say X *captures* s, D if for each infinite $Y \subseteq X$ here is initial segment t of Y s.t. $(s \cup t, X/t) \in D$.

Lemma 8 (Lemma 26.36 in Jech)**.** *For any open dense D and every infinite $Y \subseteq \omega$ and every finite $s \subseteq \omega$ there is infinite $X \subseteq Y/s$ s.t. X captures s, D.*

Proof. Inductively define $Y_0 \supseteq Y_1 \supseteq \dots$ and $m_n = minY_n$ s.t. for each $t \subseteq \{m_0 \dots m_n\}$, if there is $Y \subseteq Y_{n+1}$[3] s.t. $(s \cup t, Y) \in D$ then $(s \cup t, Y_{n+1}) \in D$. We can find Y_{n+1} by the following process: enumerate $t \subseteq \{m_0 \dots m_n\}$ as $t_0 \dots t_m$, define $Z_i \subseteq Z_{i-1}$ as the set such that $(s \cup t_i, Z_i) \in D$ if there is such Z_i, otherwise let $Z_i = Z_{i-1}$. Let $Y_{n+1} = Z_m$, then Y_{n+1} satisfies the condition.

Let $Y = \{m_n \mid n \in \omega\}$. $U = \bigcup\{[t, S] \mid (t, S) \in D\}$ is open dense in Ellentuck topology, hence it is Ramsey conull, find $X \subseteq Y$ s.t. $[s, X] \subseteq U$. We show that X captures s, D.

Let $Z \in [X]^\omega$, then $s \cup Z \in U$, hence there is (t, S) s.t. $s \cup Z \in [t, S]$, let $t' = t - s$, hence $(s \cup t', Z/t) \in D$, let $maxt = m_n$, $Z - t$ is an infinite subset of Y_{n+1} and hence by $(s \cup t, Y_{n+1}) \in D$ we have $(s \cup t, X/t) \in D$. □

Lemma 9 (Lemma 26.37 in Jech)**.** *For every infinite $Y \subseteq \omega$ there is $X \subseteq Y$ s.t. for any $s \in \omega^{<\omega}$, X/s captures s, D.*

Proof of theorem 5. Right to left: Let $x \subseteq \omega$ and assume that for every mad family $\mathcal{A} \in M$ there is $X \in \mathcal{A}$ s.t. $x - X$ is finite.

In the poset $\mathcal{P}(\omega)/fin$, ordered by almost inclusion, the above lemma entails that $\{X \in [\omega]^\omega \mid$ for any $s \in \omega^{<\omega}$, X/s captures $s, D.\}$ is dense in $\mathcal{P}(\omega)/fin$. Pick a maximal antichain from the dense set and it would be a mad family $\mathcal{A}$ satisfying for any $X \in \mathcal{A}$ $s \in \omega^{<\omega}$, X/s captures s, D.

Let $X \in \mathcal{A}$ be such that $x - X$ is finite, then $x \subseteq s \cup X$ for some initial segment s of x, as X/s captures s, D, for each infinite $Y \subseteq X$ there is initial segment t of Y s.t. $(s \cup t, X/t) \in D$.

Consider the set $W = \{t \subseteq X/s \mid (s \cup t, X/t) \notin D\}$, notice: X/s captures s, D iff W is well ordered by $t_0 \prec t_1$ iff t_1 is an initial segment of t_0.

That's because if $t_0 \prec t_1 \dots$, then for $\bigcup_i t_i$, there is $t \subseteq \bigcup_i t_i$ s.t. $(s \cup t, X/t) \in D$, hence some t_i must not satisfy $(s \cup t, X/t) \notin D$. Conversely, if Y satisfy for each $t \subseteq Y$, $(s \cup t, X/t) \notin D$, then the initial segments of Y would form an infinite descending chain.

Hence X/s captures s, D is an absolute notion, and working in V, let $Y = x/s \subseteq X$ for some initial segment s of x, then there is t s.t. $(s \cup t, x/t) \in D$, since $x \subseteq s \cup t \cup X$, the filter G_x meets D. As D is arbitrary open dense set, G_x is generic and hence x is Mathias over M.

Left to Right: For a mad family $\mathcal{A}$, the downward closure of the set $\{(s, A - s) \mid s \in [\omega]^{<\omega}, A \in \mathcal{A}\}$ is dense. Hence it intersects G_x, which means that $x \in [s, A - s]$ for some s, A, this show that $x - A = s$. □

Remark: From the proof we see the key idea of the notion capturing.

As a corollary, a subset of a Mathias real is Mathias.

3.3 Ramseyness in Solovay Model

We show the following theorem:

[3] There seems to be a typo in Jech, as discussed in https://mathoverflow.net/questions/169066/a-characterization-of-mathias-reals

Theorem 6. *Let G be generic over Levy Collapse, then $M[G] \models$ all projective sets are completely Ramsey.*

Proof. It suffice to show that for any set of reals Solovay over $M[u]$ for some $u \in Ord^{\omega}$, it is Ramsey. Let $X \subseteq \omega$ be such that

$$x \in X \iff M[u, x] \models \varphi(x)$$

For arbitrary pair (s, A), and the generic real $\dot{x}$, by Lemma 7, we obtain $H \subseteq A$ s.t. $(s, H) \models \varphi(\dot{x})$ or $(s, H) \models \neg\varphi(\dot{x})$.

Assume $(s, H) \models \varphi(\dot{x})$ is the case, let $\mathbb{P}$ be Mathias forcing in $M[u]$, then the forcing poset is of size ω and there are countable dense sets in $M[u]$ as $\aleph_1^{M[G]}$ is inaccessible in $M[u]$. Hence there is generic filter G s.t. $(s, H) \in G$.

We show that $[s, x_G] \subseteq X$, for $y \in [s, x_G]$, we have $y - x_G$ is finite and by theorem 5, y is Mathias, as $y \in [s, x_G] \subseteq [s, H]$, $(s, H) \in \varphi(\dot{x})$, hence the generic filter giving rise to y will intersect s, H and hence $M[u, y] \models \varphi(y)$. This means that $y \in X$.

The other case is similar.

□

3.4 Σ^1_2 sets and Ramseyness

Theorem 7. *If there is a Mathias real r over $L[a]$, then all $\Sigma^1_2[a]$ sets are completely Ramsey.*

Proof. We omit the parametrization.

Take X a Σ^1_2 set. Consider the name for the generic real $\dot{x}$, notice there is condition (s, A) s.t. $x \in [s, A]$ s.t. $(s, A) \Vdash X(\dot{x})$ or $(s, A) \Vdash \neg X(\dot{x})$. That's because since either $X(r)$ or $\neg X(r)$, $L[r] \models X(r)$ or $L[r] \models \neg X(r)$ by absoluteness. By forcing theorem there is condition (s, A) with $x \in [s, A]$ s.t. $(s, A) \Vdash X(\dot{x})$ or $(s, A) \Vdash \neg X(\dot{x})$.

We show that $[s, r/s] \subseteq X$ if $(s, A) \Vdash X(\dot{x})$. For $y \subseteq r$, y is also Mathias and $y \in [s, A]$, hence there is G generic over L giving rise to y and $(s, A) \in G$. $L[G] \models X(y)$ and by Shoenfield absoluteness $X(y)$.

The other side is similar. □

Remark: the special property of Mathias forcing allows us to obtain the result that is slightly stronger than ordinary focing.

This style of argument gives another proof of the classical results:

Theorem 8. *All analytic sets are completely Ramsey*

Proof. For an analytic property φ and a condition (s, A), pick a ctm of enough ZFC s.t. $(s, A) \in M$. Working in M, there is $(s, B) \leq (s, A)$ s.t. $(s, B) \Vdash \varphi(\dot{x})$ or $(s, B) \Vdash \neg\varphi(\dot{x})$ by pure decision. w.l.o.g. assume $(s, B) \Vdash \varphi(\dot{x})$.

As M is countable, pick generic filter G containing (s, B), we show that $[s, x_G/s] \subseteq \{x \mid \varphi(x)\}$. For $y \in [s, x_G/s] \subseteq [s, B] \subseteq [s, A]$, since y is also Mathias, there is generic filter H corresponding to y. As $(s, B) \in H$, $M[H] \models \varphi(y)$, by Mostowski absoluteness of analytic properties, $\varphi(y)$. □

Theorem 9. *All analytic sets are Lebesgue measurable and have the Baire property.*

Proof 1, the proof modeling Judah-Shelah. We only show the Lebesgue measurable part. For an analytic property φ, pick an ctm of enough ZFC and consider the set

$$D = \{A_c^M \in (\mathcal{B}/I_n)^M \mid A_c^M \Vdash \varphi(\dot{x}) \text{ or } A_c^M \Vdash \neg\varphi(\dot{x})\}$$

Here $\dot{x}$ is the name of generic real and D is dense in the forcing condition $(\mathcal{B}/I_n)^M$ in M. Pick an antichain W of this dense set and consider

$$\begin{aligned}
Y_0 &= \{c \in BC^M \mid A_c^M \in W, A_c^M \Vdash \neg\varphi(\dot{x})\} \\
Y_1 &= \{c \in BC^M \mid A_c^M \in W, A_c^M \Vdash \varphi(\dot{x})\} \\
Z_0 &= \bigcup\{A_c \mid c \in Y_0\} \\
Z_1 &= \bigcup\{A_c \mid c \in Y_1\}
\end{aligned}$$

It then suffice to show that $Z_0-\varphi$ and $Z_1-\neg\varphi$ is null as then as $Z_0 \cap Z_1$ is null and $\mu^{L[a]}(Z_0)+\mu^{L[a]}(Z_1) = 1$ by maximality of W, hence $\varphi - Z_0 \approx \varphi \cap Z_1 = Z_1 - \neg\varphi$, here $\approx$ means equal in measure. This means that $Z_0 \Delta \varphi$ is null, and hence since Z_Q is a countable union of Borel sets, A is Lebesgue measurable.

To show $Z_0 - \varphi$ is null, notice $Z_0 \cap R(M) \subseteq \varphi$ as if x is random over M and $x \in A^c$ where $A_c^M \Vdash \varphi(\dot{x})$, we know there is generic G s.t. $M[G] \models \varphi(x)$ and by Mostowski absoluteness $\varphi(x)$. Hence $Z_0 - \varphi$ are reals that are not random over M, thus countable. Hence $Z_0 - \varphi$ is null. $\square$

Proof 2, the proof modeling Solovay. Let φ be an analytic set and $T \in \omega \times \omega$ a tree such that $p[T] = \varphi$. Let M be a ctm with enough ZFC containing T and we have

$$\begin{aligned}
\varphi(x) &\iff T(x) \text{ is not well founded} \\
&\iff M[x] \models T(x) \text{ is not well founded}
\end{aligned}$$

by absoluteness of well foundedness. Hence $\varphi(x)$ is Solovay over M.

As reals random over M are co-null, by Corollary 26.6 of Jech $\varphi(x)$ is Lebesgue measurable. $\square$

Remark: This kind of argument are precisely the Solovay style argument in analysis of Σ_2^1 sets. The key difference between the analytic case and Σ_2^1 case is that Mostowski absoluteness allows us to work in an ctm, which naturally has co-null many random reals. (My proof does not explicitly use the Mostowski absoluteness, but that's what's the core of my argument.)

4 Ramsey Property implies no mad Family

4.1 Proof in the paper [ST19]

Given that R-uniformization is true, we show the non-existence of Ramsey property:

Definition 3 (R-Uniformization)**.** *For all Y Polish and all $R \subseteq [\omega]^\omega \times Y$ with full projection, there is an infinite set $A \subseteq \omega$ and a function $\theta : [A]^\omega \to Y$ which uniformizes R on $[A]^\omega$.*

Now we study the proof idea in the paper The Ramsey property implies no mad families.

Theorem 10 (ZF + DC + R-Unif)**.** *If all sets have the Ramsey property, then there are no infinite mad families.*

The chief idea of this proof is the following:
(1) For the first part we define a function

$$T : [\omega]^\omega \to 2^{\omega^{<\omega}}$$
$$z \mapsto T^z$$

where the map can be thought of mapping to a tree of approximation of uniformizing functions of of set $\{(z, a) \mid a \in \mathcal{A}, |a \cap \tilde{z}| = \infty\}$. Here $\tilde{z}$ is a scattered correspondence of z. The map is E_0 invariant.

(2) By the fact that all sets are Ramsey, we use the following proposition 2 and get a $w \in [\omega]^\omega$ s.t. $T|_{[w]^\omega}$ is continous, thus constant as T is E_0 invariant.

This leads to a contradiction.

Proposition 2 (Proposition 2 in [ST19])**.** *Assuming all sets are Ramsey, $\theta : [\omega]^\omega \to Y$ is a function where Y is a separable space, then there is $w \in [\omega]^\omega$ s.t. $\theta|_{[w]^\omega}$ is continous w.r.t. the standard topology on $[\omega]^\omega$.*

Note: For reasonable point class Γ, every sets in Γ is Ramsey iff all Γ sets are completely Ramsey.

Proof. Enumerate the basis of Y as $U_0, U_1 \dots$, and consider $\theta^{-1}[U_0], \theta^{-1}[U_1], \dots$. These sets are completely Ramsey by assumption. Then we inductively pick $x_0 \supseteq x_1 \supseteq \dots$ infinite subset of ω, $m_n = min x_n$ s.t. for all $s \subseteq \{m_0 \dots m_n\}$, either $[s, x_n] \subseteq \theta^{-1}[U_n]$ or $[s, x_n] \cap \theta^{-1}[U_n] = \emptyset$. Take $x = \{min x_n \mid n \in \omega\}$ and we show that $\theta|_{[x]^\omega}$ is continuous w.r.t. the standard topology on $[\omega]^\omega$.

Then $\theta|_{[x]^\omega}^{-1}[U_n] = \bigcup\{N_s \subseteq [x]^\omega \mid s \subseteq \{m_0 \dots m_n\}$ and $[s, x_n] \subseteq \theta^{-1}[U_n]\}$, here N_s denotes the basic subset $N_s = \{y \in [x]^\omega \mid s \subseteq y\}$. □

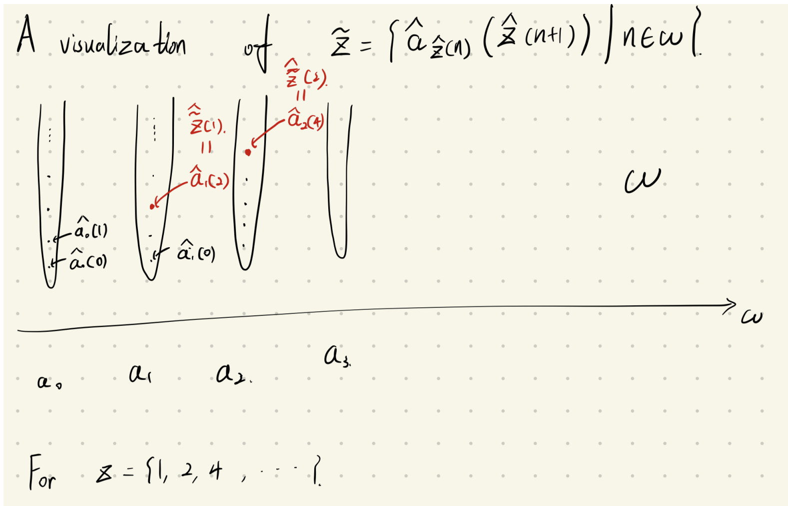
Proof of theorem 10. Assume for contradiction that $\mathcal{A}$ is an infinite mad family.

Step 1: Define the scattered verion of z.

Let $a_n, n \in \omega$ be an sequence of elements in $\mathcal{A}$. We can assume without loss of generality that $a_n \cap a_m = \emptyset$ if $n \neq m$. For an infinite set x of ω, let $\widehat{x}$ be the unique increasing funciton enumerating it. For each $z \in [\omega]^\omega$, we define the following element:

$$\tilde{z} = \{\widehat{a_{\widehat{z}(n)}}(\widehat{z}(n+1)) \mid n \in \omega\}$$

Here is a visualization of the set $\tilde{z}$:



The idea is to transform z into a very scattered set, w.r.t. a_n. Here are some basic observation of the transformation:

1. For each a_n, $|a_n \cap \tilde{z}| \leq 1$.
2. If $z_1 \Delta z_2$ is finite, then $\tilde{z_1} \Delta \tilde{z_2}$ is finite.

The second bullet point is because if $z_1 \Delta z_2$, then (w.l.o.g.) there is $k \in \omega$ and $N \in \omega$ s.t. for all $n > N$, $\widehat{z_1}(n+k) = \widehat{z_2}(n)$ and thus $a_{\widehat{z_1(n+k)}}(\widehat{z_1}(n+k+1)) = \widehat{a_{\widehat{z_2}(n)}}(\widehat{z_2}(n+1))$.

Step 2: Define the map $z \mapsto T^z$

Given $\mathcal{A}$ is an infinite mad family, the set

$$R = \{(z, a) \in [\omega]^\omega \times \mathcal{A} \mid |\tilde{z} \cap a| = \infty\}$$

has full projection. Indeed notice by the property of $\tilde{z}$, the projection of this set along $\mathcal{A}$ is indeed in $\mathcal{A} - \{a_n \mid n \in \omega\}$.

Apply R-Unif and we obtain $W \in [\omega]^\omega$ and $\theta : [W]^\omega \to \mathcal{A} - \{a_n \mid n \in \omega\}$ s.t. $\tilde{z} \cap \theta(z)$ is infinite. Apply proposition 2 and we obtain $\theta|_{[W_0]^\omega}$ is continuous for some $W_0 \in [W]^\omega$. Then $\mathcal{B} = \theta[[W_0]^\omega]$ is analytic and hence there is tree T on $2 \times \omega$ s.t. $p[T] = \mathcal{B}$.

For each $z \in [W_0]^\omega$, consider

$$T^z = \{t \in T \mid \exists y \in p[T_t], |y \cap \tilde{z}| = \infty\}$$

Here, an element in $p[T_t]$ should of course be a sequence of elements 2, we identify it with a subset of ω.

Step 3: The map $\mathbb{T} : z \mapsto T^z$ is constant on a large set

First observe that T^z is E_0 invariant, meaning that if $z_1 \Delta z_2$ is finite then $T^{z_1} = T^{z_2}$. That's because if $z_1 \Delta z_2$ is finite, then by second observation of $\tilde{z}$, $\tilde{z_1} \Delta \tilde{z_2}$ is finite and hence $|y \cap \tilde{z_1}| = \infty$ iff $|y \cap \tilde{z_2}| = \infty$. Hence $T^{z_1} = T^{z_2}$.

Then apply 2 and we get $W_1 \in [W_0]^\omega$ s.t. the map $\mathbb{T} : z \mapsto T^z$ is continuous on $[W_1]^\omega$, w.r.t. the standard topology. This means that the map is constant given the map is E_0 invariant:

First notice for any basic open $N_s, N_{s'} \subseteq [W_0]^\omega$, we have $\mathbb{T}[N_s] = \mathbb{T}[N_{s'}]$, that's because for any $z \in N_s$ there is $z' \in N_{s'}$ s.t. zE_0z' and hence $T^z = T^{z'}$. For $z \in [W_0]^\omega$, $\mathbb{T}[\{z\}] = \bigcap\{\mathbb{T}[N_s] \mid s \subseteq z\}$, since the right is a singleton, each $\mathbb{T}[N_s]$ is and hence $\mathbb{T}[[W_0]^\omega] = \mathbb{T}[N_\emptyset]$ is a singleton.

Step 4: An analysis of the tree T^* that leads to a contradiction

Let $T^z = T^*$ for $z \in [W_1]^\omega$.

Lemma 10 (Subclaim 3)**.** *If there is $t_0, t_1 \in T^*$ and n s.t. for any $y_0 \in p[T_{t_0}]$, $y_1 \in p[T_{t_1}]$, $n \in y_0 \Delta y_1$, then there is node $s_0 \supseteq t_0$ and $s_1 \supseteq t_1$ and $k \in \omega$ s.t. for any infinite branches $y_0 \in p[T_{s_0}]$, $y_1 \in p[T_{s_1}]$, there intersection is bounded by k: $y_0 \cap y_1 \subseteq k$.*

Proof. Assume for contradiction that it is not the case. Then for any node $t_0' \supseteq t_0$, $t_1' \supseteq t_1$ and $k \in \omega$ we can find branches $y_0 \in p[T_{t_0'}]$ and $y_1 \in p[T_{t_1'}]$ and $m \geq k$ s.t. $m \in y_0 \cap y_1$.

Hence we know that $y_0(m) = y_1(m) = 1$ and let $t_0'' = (y_0|_m, s_0) \in T^*$ and $t_1'' = (y_1|_m, s_1) \in T^*$, we know that for any branches y_0', y_1' in $p[T_{t_0''}], p[T_{t_1''}]$, $m \in y_0' \cap y_1'$.

Proceeding inductively, we can find nodes $t_0 \subseteq u_0 \subseteq u_1 \dots$ and $t_1 \subseteq v_0 \subseteq v_1 \dots$ and $m_0 < m_1 \dots$ s.t. for any branches y_0, y_1 extending u_i, v_i, $m_i \in y_0 \cap y_1$. This means that for u, v, the corresponding projection of $\bigcup_i u_i, \bigcup_i v_i$, we have $\{m_i \mid i \in \omega\} \subseteq u \cap v$ and $n \in u \Delta v$, this is a contradiction to the fact that $u, v \in \mathcal{B} \subseteq \mathcal{A}$, they are elements of a mad family. ☐

Then we claim that the following result holds true: $|p[T^*]| \leq 1$. The argument essentially uses the Ramsey largeness of the constant set $[W_0]^\omega$.

Assume for contradiction that $y_0 \neq y_1 \in p[T^*]$, then let t_1, t_2 be two nodes at which y_0, y_1 disagree at the first coordinate, and hence by the above lemme we find s_0, s_1 and k s.t. for any y_0, y_1 in $p[T_{s_0}], p[T_{s_1}]$ respectively, $y_0 \cap y_1 \in k$.

Let $a_n^i = a_n \cap \bigcup\{y \mid y \in p[T_{s_i}]\}$ for $i = 0, 1$, then $a_n^0 \cap a_n^1 \subseteq k$ for all n. By removing an initial segment from W_0, we may assume that $a_n^0 \cap a_n^1 = \emptyset$.

For each n, it is either the case that:

1) $\exists^\infty j \in W_0, \widehat{a_n}(j) \in a_n^0$.

2) $\exists^\infty j \in W_0, \widehat{a_n}(j) \notin a_n^0$.

Inductively pick infinite W_0^n so that $W_0^n \subseteq W_0^{n-1}$ and for each $j \in W_0^n$, $\widehat{a_n}(j) \in a_n^0$ or for each $j \in W_0^n$, $\widehat{a_n}(j) \notin a_n^0$. Hence let $W_1 = \{min(W_0^n) \mid n \in \omega\}$, and this set satisfies for each n, it is either the case that:

1) $\forall j \in W_1 - n, \widehat{a_n}(j) \in a_n^0$.

2) $\forall j \in W_1 - n, \widehat{a_n}(j) \notin a_n^0$.

Finally, let $W_2^0 = \{n \in W_1 \mid \forall j \in W_1 - n, \widehat{a_n}(j) \in a_n^0\}$ and $W_2^1 = \{n \in W_1 \mid \forall j \in W_1 - n, \widehat{a_n}(j) \notin a_n^0\}$, one of them must be infinite. Let W_2 be this infinite set, then it is either the case that:

1) $\forall n \in W_2 \forall j \in W_2 - n, \widehat{a_n}(j) \in a_n^0$.

2) $\forall n \in W_2 \forall j \in W_2 - n, \widehat{a_n}(j) \notin a_n^0$.

This leads to a contradiction as if the former, then $z \in [W_2]^\omega$ satisfies $\tilde{z} \cap \bigcup p[T_{s_1}] = \emptyset$, contradicting the fact that $s_1 \in T^* = T^z$. If the latter, $\tilde{z} \cap \bigcup p[T_{s_0}] = \emptyset$ and the similar contradiction is reached.

Finally, we show that indeed $p[T^*] = \emptyset$, which is contradictory as at least $\theta(z)$ should be in $p[T^z] = p[T^*]$. To show $p[T^*] = \emptyset$, for $\theta(z)$, we find another z' s.t. $\tilde{z'} \cap \theta(z) = \emptyset$. Take

$$\widehat{a_{\widehat{z'}(n)}}(\widehat{z'}(n+1)) > max(a_{\tilde{z'}} \cap \theta(z))$$

and we are done. $\square$

4.2 Localizing the Proof

Definition 4 (R-Uniformization for Γ, Γ')**.** *For all Y Polish and all Γ set $R \subseteq [\omega]^\omega \times Y$ with full projection, there is an infinite set $A \subseteq \omega$ and a Γ' measurable function $\theta : [A]^\omega \to Y$ which uniformizes R on $[A]^\omega$.*

Proposition 3. *Assuming $\theta : [\omega]^\omega \to Y$ is a Γ measurable function where Y is a separable space, and all Γ sets are completely Ramsey then there is $x \in [\omega]^\omega$ s.t. $\theta|_{[x]^\omega}$ is continous w.r.t. the standard topology on $[\omega]^\omega$.*

Theorem 11. *Assume that Γ, Γ' are among $\mathbf{\Sigma}^1_n, \mathbf{\Pi}^1_n$ or $\mathbf{\Delta}^1_n$ and the following:*

1. *R-Uniformization for Γ, Γ'.*
2. *All Γ' sets are completely Ramsey.*

Then there are no Γ mad families.

Proof sketch. We analyze where 3 and R-uniformization are used. By the fact that the map $\mathbb{T}$ is $\sigma(\mathbf{\Sigma}^1_1)$ measurable by their talk, the conclusion follows. $\square$

Remark: The theorem does not establish that there are no $\mathbf{\Pi}^1_1$ mad family, even though we have Kondo's uniformization and all $\mathbf{\Pi}^1_1$ sets are completely Ramsey. That's because Kondo's uniformization does not give us a $\mathbf{\Pi}^1_1$ measurable function , but rather a $\mathbf{\Pi}^1_1$ function.

For a $\mathbf{\Pi}^1_1$ function $f : X \to \mathcal{N}$, and basic open U,

$$x \in f^{-1}(U) \iff \exists^{\mathcal{N}} y(y \in U \wedge (x, y) \in f) \iff \forall^{\mathcal{N}} y((x, y) \in f \to y \in U)$$

This just means that f is $\mathbf{\Delta}^1_2$ measurable, but not $\mathbf{\Pi}^1_1$ measurable. Hence the conclusion does not follow.

Let's state the local theorem in the terms of point class belonging:

Corollary 1. 1. *If all $\mathbf{\Delta}^1_n$ sets are completely Ramsey and all relations $\mathcal{R} \subseteq [\omega]^\omega \times \mathcal{N}$ in point class Γ can be uniformized by a $\mathbf{\Sigma}^1_n$ function, then there are no Γ mad family.*

2. *If all $\mathbf{\Delta}^1_{n+1}$ sets are completely Ramsey and all relations $\mathcal{R} \subseteq [\omega]^\omega \times \mathcal{N}$ in point class Γ can be uniformized by a $\mathbf{\Pi}^1_n$ function, then there are no Γ mad family.*

Proof. All $\mathbf{\Pi}^1_n$ function is $\mathbf{\Delta}^1_{n+1}$ measurable and all $\mathbf{\Sigma}^1_n$ function is $\mathbf{\Delta}^1_n$ measurable. $\square$

Corollary 2. *If there is a Mathias real over $L[a]$, then all $\Sigma^1_2[a]$ sets are not mad.*

Proof. By theorem 7, Kondo uniformization and the localized theorem. $\square$

Note: I assume that Kondo uniformization gives $\Sigma^1_2[a]$ uniformizing function for $\Sigma^1_2[a]$ relations.

Question: For what kind of point classes point class Γ, every sets in Γ is Ramsey iff all Γ sets are completely Ramsey?

5 Tree Analysis of mad Family

We study the tree analysis of mad family in the [Tör18] paper.

A notation: For node $t \in T$, where T is a tree on $2 \times \omega$, we write $t = (t_*, t^*)$ i.e. t_* denotes the first indice and t^* the second.

Definition 5 (Definition 2.3 in [Tör18])**.** *For a tree T on $2 \times \omega$, $\alpha \leq \omega$, a sequence $t^j_i \in T$, $i < \alpha$ $j \in \{0, 1\}$ is a diagonal sequence in T of length α if:*

1. $h(t^0_i) = h(t^1_i)$.
2. $t^1_i \subseteq t^j_{i+1}$
3. *t^0_i and t^1_i are incompatible in the first coordinate*
4. *For all $s, t \in T$ with $s \supseteq t^0_i$ and $t \supseteq t^1_i$ we have $s_* \cap t_* = (t^0_i)_* \cap (t^1_i)_*$*
5. *$p[T_{t^0_i}] \neq \emptyset$ for all $i \leq \alpha$.*

The key condition is perhaps the forth condition.

A finite T diagonal sequence is T-terminal iff it cannot be further extended. A top leaf of a finite T diagonal sequence of length n is the node t^1_n.

$$\tau(T) = \{t \in T \mid t \text{ is top node of some } T\text{-terminal diagonal sequence.}\}$$

Lemma 11 (Lemma 2.4. in [Tör18])**.** *A tree T of $2 \times \omega$ s.t. $p[T]$ is an uncountable almost disjoint family admits an infinite diagonal sequence.*

Proof sketch. The proof is basically the tree pruning argument for perfect set properties.

First, notice by the same argument in Lemma 10, for a T-terminal finite diagonal sequence with t^1_n as its terminal node, $p[T_{t^1_n}]$ has at most one element.

Then, define the pruning process as

$$T^{\alpha+1} = T^{\alpha} - \{t \in T^{\alpha} \mid \exists s \in \tau(T^{\alpha}), t \supseteq s\}$$

If T has no infinite diagonal sequence, there will be α s.t. $T^{\alpha} = \emptyset$, α is countable. Then $p[T] = \bigcup_{\beta<\alpha} \bigcup_{t\in\tau(T^{\beta})} p[T^{\beta}_t]$, which is a countable union of singletons. $\square$

5.1 All analytic sets are not mad.

With this techique we can show that all analytic sets are not mad.

Theorem 12 (Mathias)**.** *There is no analytic mad family.*

We need the following Lemma:

Lemma 12 (Lemma 2.2 in [Tör18])**.** *For any family $\mathcal{A}$ of subset of ω, if there is a sequence $A_n \subseteq \omega$ for $n \in \omega$ s.t.*

1. *Every $x \in \mathcal{A}$ is almost contained in a union of finitely many A_n.*
2. *For all n, $\omega - \bigcup_{i\leq n} A_i$ is infinite.*

Then there is an infinite $z \subseteq \omega$ s.t. $z \cap x$ is finite for all $x \in \mathcal{A}$.

Proof. Take m_n as the least element in $(\omega - \bigcup_{i\leq n} A_i)$ greater than m_{n-1}, then $\{m_n \mid n \in \omega\}$ works. $\square$

Proof sketch of theorem 12. The argument is to kill all the infinite diagonal sequences in the tree and at the mean time create a cover that satisfies the condtion of the above lemma.

Let $\mathcal{A}$ be an analytic almost disjoint family, we show that it is not maximal. Let $\mathcal{A} = p[T]$, T is a tree on $2 \times \omega$.

For each ordinal α until some α^*, let T^{α} be a subtree of T, $t^i_j(\alpha)$ be an infinite diagonal sequence in T^{α} and $A^{\alpha}_i \subseteq \omega$ for $i \in \omega$.

The following are the key definitions:

1. $A^{\alpha}_i = \bigcup p[T^{\alpha}_{t^0_i(\alpha)}]$
2. $T^{\alpha+1} = \{t \in T^{\alpha} \mid \exists x \in p[T^{\alpha}_t], \forall k \in \omega, \forall \alpha_1 \dots \alpha_k < \alpha, \forall i_1 \dots i_k \in \omega, x \not\subseteq^* \bigcup_{j<k} A^{\alpha_j}_{i_j}\}$
3. The sequence terminates when T^{α} no longer has any infinite diagonal sequences. Then, the terminating ordinal is α^*.

Since $t^0_i(\alpha) \notin T^{\alpha+1}$, each step there are countably many nodes pruned, given T is countable, the process terminates at $\alpha^* < \omega_1$. Then, the set $\{A^{\alpha}_i \mid \alpha < \alpha^*, i \in \omega\}$ is countable.

Since $p[T^{\alpha^*}] \subseteq p[T]$ and is an almost disjoint family, and it does not contain an infinite diagonal sequence, by 11 it is at most countable. Let A_i enumerate

$p[T^{\alpha^*}] \cup \{A_j^\alpha \mid j \in \omega, \alpha < \alpha^*\}$. We show that A_i satisfies the condition in the previous lemma.

Condition 1: if $x \in p[T^{\alpha^*}]$ we are done and if not, it holds true by property 2 of the sequence.

Condition 2 is verified using the property 4 of diagonal sequence.

□

Corollary 3. *Every analytic a.d. family is contained in a proper ideal generated by a countable family of subsets of $\mathcal{A}$.*

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